

pc-imaging-server

Automated Windows 10/11 deployment server for Juniper Design. PXE boots desktops and laptops from the network, runs unattended installs, applies all Windows updates, installs required software, configures printer queues, joins Wi-Fi, and removes bloatware — zero-touch from power-on to ready desk.

Built at [Juniper Design](#).

What it does

Phase	Script	Description
0	<code>00-rename-server.ps1</code>	Rename the imaging server host to <code>pc-deploy</code>
1	<code>01-setup-wds.ps1</code>	Install WDS + configure PXE boot service
2	<code>02-setup-dhcp-options.ps1</code>	Ubiquiti router DHCP option 66/67 configuration
3	<code>03-windows-update.ps1</code>	Force-apply all Windows updates on a target PC
4	<code>04-install-packages.ps1</code>	Install MSIs and winget packages
5	<code>05-setup-printers.ps1</code>	Add and share printer queues
6	<code>06-join-wifi.ps1</code>	Join a target PC to the office Wi-Fi profile
7	<code>07-remove-bloatware.ps1</code>	Remove unwanted apps and disable junk Windows features

Server requirements

- Windows Server 2019/2022 or Windows 10/11 Pro (WDS role)
- Static IP on the LAN segment used for PXE
- Ubiquiti UniFi or EdgeRouter for DHCP options 66/67
- Remote Desktop enabled (pre-configured on this machine)
- IPassword CLI (`op`) installed for secret retrieval

Quick start

1 — Rename the server

```
.\scripts\00-rename-server.ps1
# Reboots automatically; reconnect via RDP to DESKTOP-I8UM43L
# after it comes back up as pc-deploy
```

2 — Set up WDS + PXE

```
.\scripts\01-setup-wds.ps1
```

Then configure DHCP — see `scripts\02-setup-dhcp-options.ps1` for the exact Ubiquiti commands to add option 66 (TFTP server IP) and option 67 (`boot\x64\wdsnbp.com`).

3 — Import Windows images

Once WDS is running, add boot and install images via WDS console or:

```
# Mount your Windows ISO first, then:
wdsutil /Add-Image /ImageFile:"D:\sources\boot.wim" /ImageType:Boot
wdsutil /Add-Image /ImageFile:"D:\sources\install.wim" /ImageType:Install /ImageGroup:"Windows"
```

4 — Drop answer files

Copy `unattend\unattend-win10.xml` or `unattend\unattend-win11.xml` into WDS console → Properties → Client → "Unattend file" for the relevant image, or embed them in the WIM with DISM.

5 — Post-install scripts

After Windows lands, run the remaining scripts in order (3 → 7) on the target PC — either manually, via a GPO startup script, or called from the unattend `FirstLogonCommands` section pointing at a UNC share on `pc-deploy`.

Ubiquiti DHCP setup

The router lives at `192.168.0.1`. Before PXE works you need to add two DHCP options on the LAN scope. Run:

```
.\scripts\02-setup-dhcp-options.ps1
```

...or see the comments inside that file for the equivalent Ubiquiti CLI / UniFi web UI steps.

Credentials

All passwords and Wi-Fi PSKs are stored in 1Password. Scripts reference them via `op run` or `op read` — **never hard-code secrets here**. The 1Password CLI must be signed in before running any script that calls `op`.

Repo structure

```
pc-imaging-server/
├─ assets/           ← Juniper brand assets (banner SVG)
├─ docs/pdf/        ← Auto-generated branded PDF docs
├─ scripts/
│   ├─ 00-rename-server.ps1
│   ├─ 01-setup-wds.ps1
│   ├─ 02-setup-dhcp-options.ps1
│   ├─ 03-windows-update.ps1
│   ├─ 04-install-packages.ps1
│   ├─ 05-setup-printers.ps1
│   ├─ 06-join-wifi.ps1
│   └─ 07-remove-bloatware.ps1
├─ unattend/
│   ├─ unattend-win10.xml
│   └─ unattend-win11.xml
├─ CLAUDE.md
├─ push.ps1
└─ README.md
```

Juniper Design internal tooling — private repository.